Sprint 01 Assignment: Child Undernutrition in Nigeria

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Dataset: Nigeria DHS 2024 Children's Recode − NGKR8BFL

Total rows: 27783
Valid WAZ observations: 9513
Valid WHZ observations: 9464
Missing WAZ (%): 65.8
Missing WHZ (%): 65.9

Breastfeeding recode follows the assignment PDF: 93 = Never, 94 = Still breastfeeding, 95 = Stopped.

Dataset Understanding, Assumptions, and Limitations

Research−quality checks from the assignment draft

Unit of analysis: one row represents one child in the Nigeria DHS 2024 Children's Recode file. Raw DHS anthropometry variables are preserved; cleaned HAZ, WAZ, and WHZ are created separately. DHS anthropometry codes 9996 and above are treated as missing before dividing valid scores by 100. Religion was checked against the DHS labels and collapsed to Christianity, Islam, and Other. The official assignment lists 15 WAZ/WHZ variables; hw70 is read only to reproduce the required HAZ comparison table. v005 is normalized and used for descriptive prevalence estimates, but no full complex−survey design adjustment is applied. The draft guidance notes hw73 plausibility flags; hw73 is not applied because it is outside the official assignment variable list.

Task B Tables and Interpretations

Table 1. Missing value table

Variable	Missing_N	Missing_Percent
haz	18373	66.13
whz	18319	65.94
waz	18270	65.76
age_months	17869	64.32
breastfeeding	11790	42.44
father_educ	1727	6.22
child_alive	0	0.00
household_size	0	0.00
mother_educ	0	0.00
region	0	0.00
religion	0	0.00
residence	0	0.00
sample_weight	0	0.00
sex	0	0.00
under5_children	0	0.00
wealth	0	0.00

Breastfeeding has the largest missing share because the assignment keeps only codes 93, 94, and 95. WAZ and WHZ have 65.8% and 65.9% missing respectively after DHS invalid anthropometry codes are removed.

Table 2. Summary statistics for WAZ and WHZ

Outcome	Mean	Median	SD	Min	Max	Skewness
WAZ	−1.263	−1.24	1.234	−5.95	4.7	−0.128
WHZ	−0.506	−0.49	1.115	−4.98	4.8	−0.125

The average WAZ is −1.26 and the average WHZ is −0.51, so weight−for−age is lower nationally than weight−for−height. Both distributions include children below the clinical −2 threshold, making prevalence estimates necessary.

Table 3. National prevalence

Indicator	Valid_N	Percent
Underweight (WAZ < −2)	9513	26.41
Severe underweight (WAZ < −3)	9513	8.27
Wasting (WHZ < −2)	9464	8.44
Severe wasting (WHZ < −3)	9464	1.87

Nationally, 26.4% of children are underweight and 8.4% are wasted. Severe underweight is 8.3%, while severe wasting is 1.9%, showing the most extreme WHZ deficits are less common.

Task B Continued

Table 4. HAZ, WAZ, and WHZ comparison

Indicator	Outcome	Valid_N	Percent
Stunting (HAZ < −2)	HAZ	9410	39.14
Underweight (WAZ < −2)	WAZ	9513	26.41
Wasting (WHZ < −2)	WHZ	9464	8.44

Stunting (HAZ < −2) is the most common of the three anthropometric deficits in this dataset. This comparison places the new WAZ and WHZ assignment results beside the HAZ stunting measure from class.

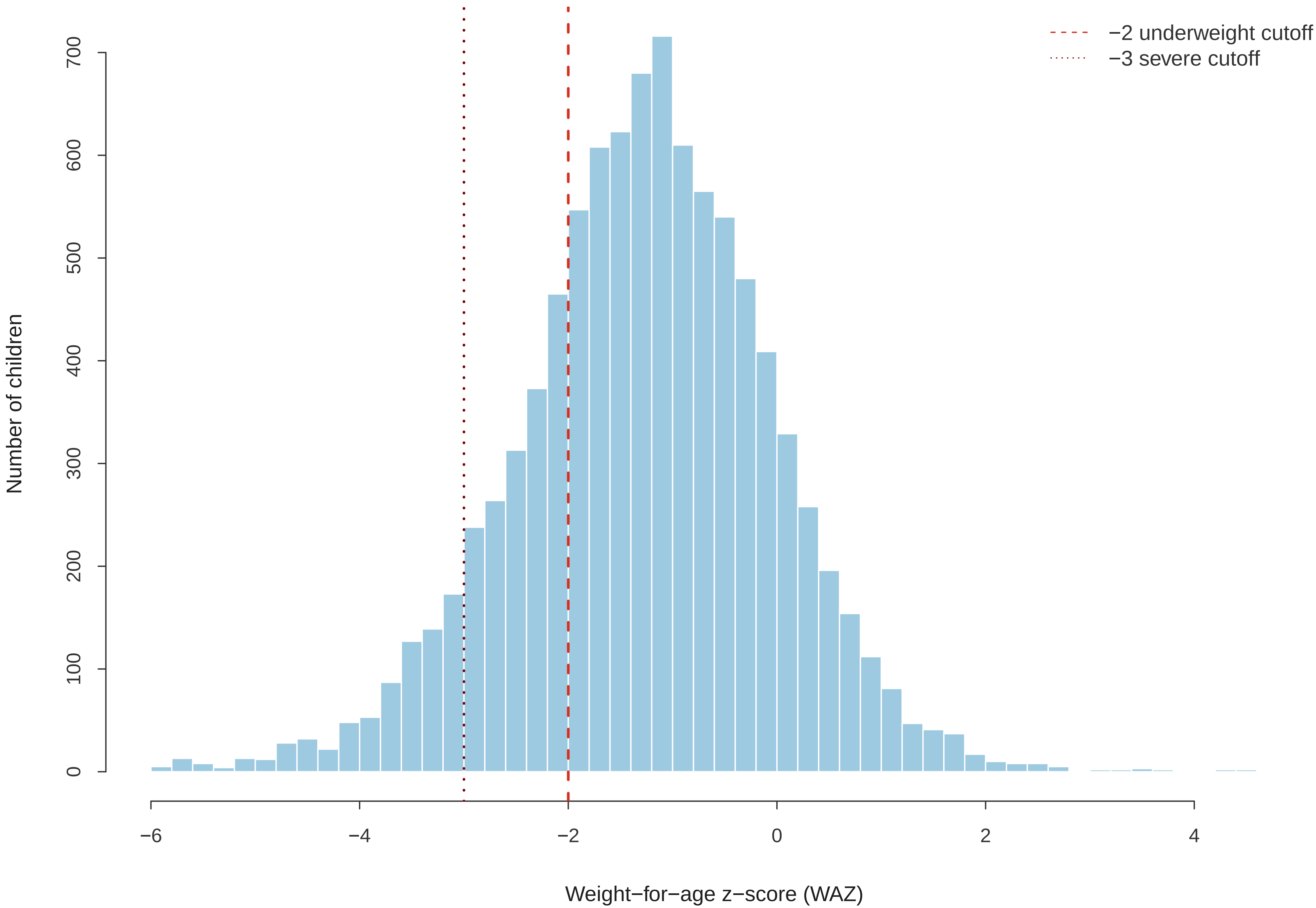
Table 5. Underweight prevalence by subgroup

Grouping	Category	Valid_N	Underweight_Percent
Region	North west	2418	33.29
Region	North east	1700	33.10
Region	North central	1690	21.35
Region	South east	1703	15.85
Region	South south	1112	17.13
Region	South west	890	21.32
Residence	Urban	4035	21.47
Residence	Rural	5478	29.99
Wealth quintile	Poorest	2078	38.60
Wealth quintile	Poorer	1734	32.05
Wealth quintile	Middle	1867	25.11
Wealth quintile	Richer	2085	21.96
Wealth quintile	Richest	1749	12.18
Mother's education	None	3524	36.11
Mother's education	Primary	1157	26.13
Mother's education	Secondary	3628	20.84
Mother's education	Higher	1204	9.79
Religion	Christianity	4340	16.38
Religion	Islam	5112	32.39
Religion	Other	61	30.01

The highest underweight prevalence appears in Poorest within Wealth quintile at 38.6%. This subgroup is the clearest priority for follow−up modelling and targeted public−health interpretation.

Figure 1. Distribution of WAZ

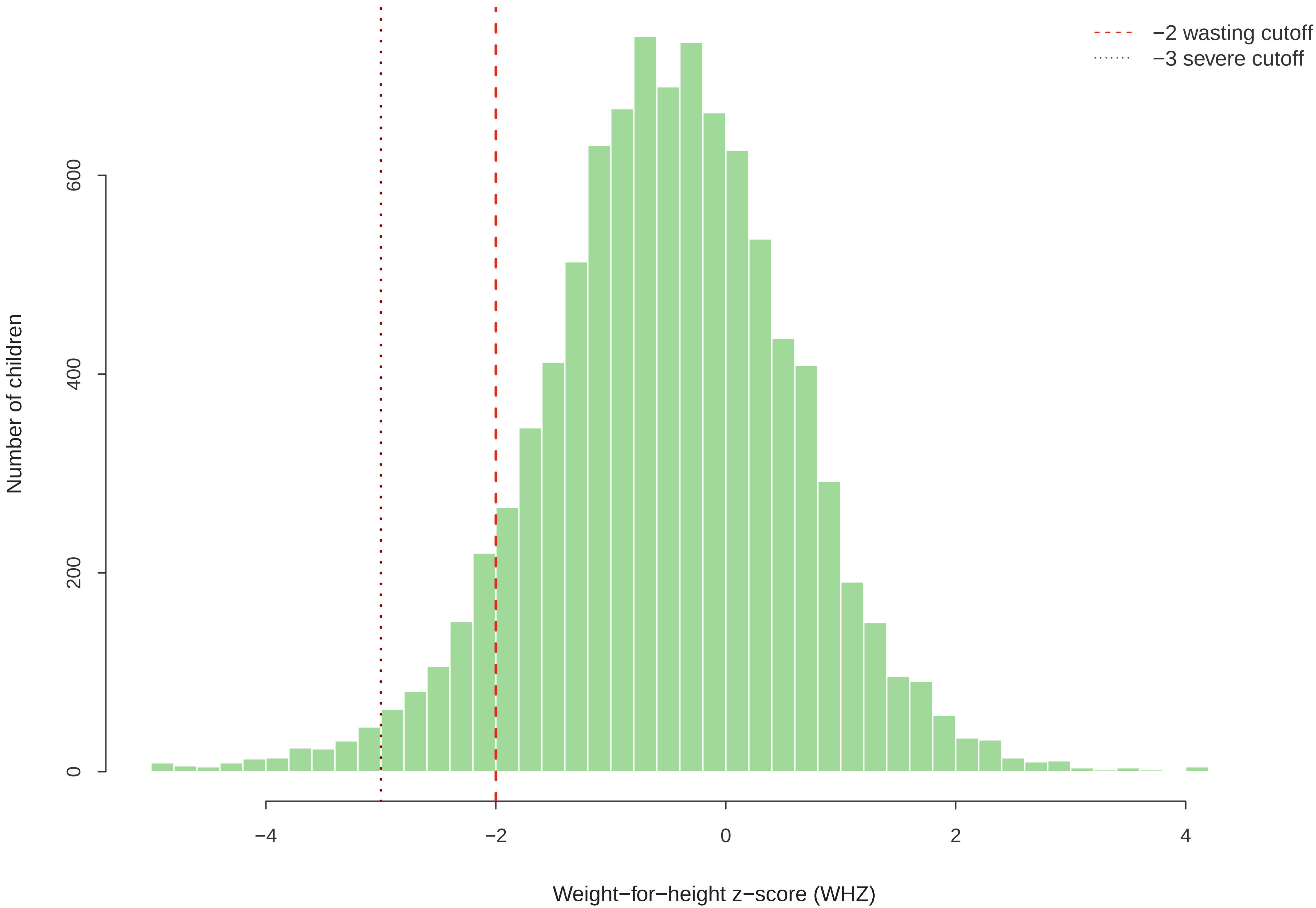
Nigeria NDHS 2024; underweight prevalence = 26.4%



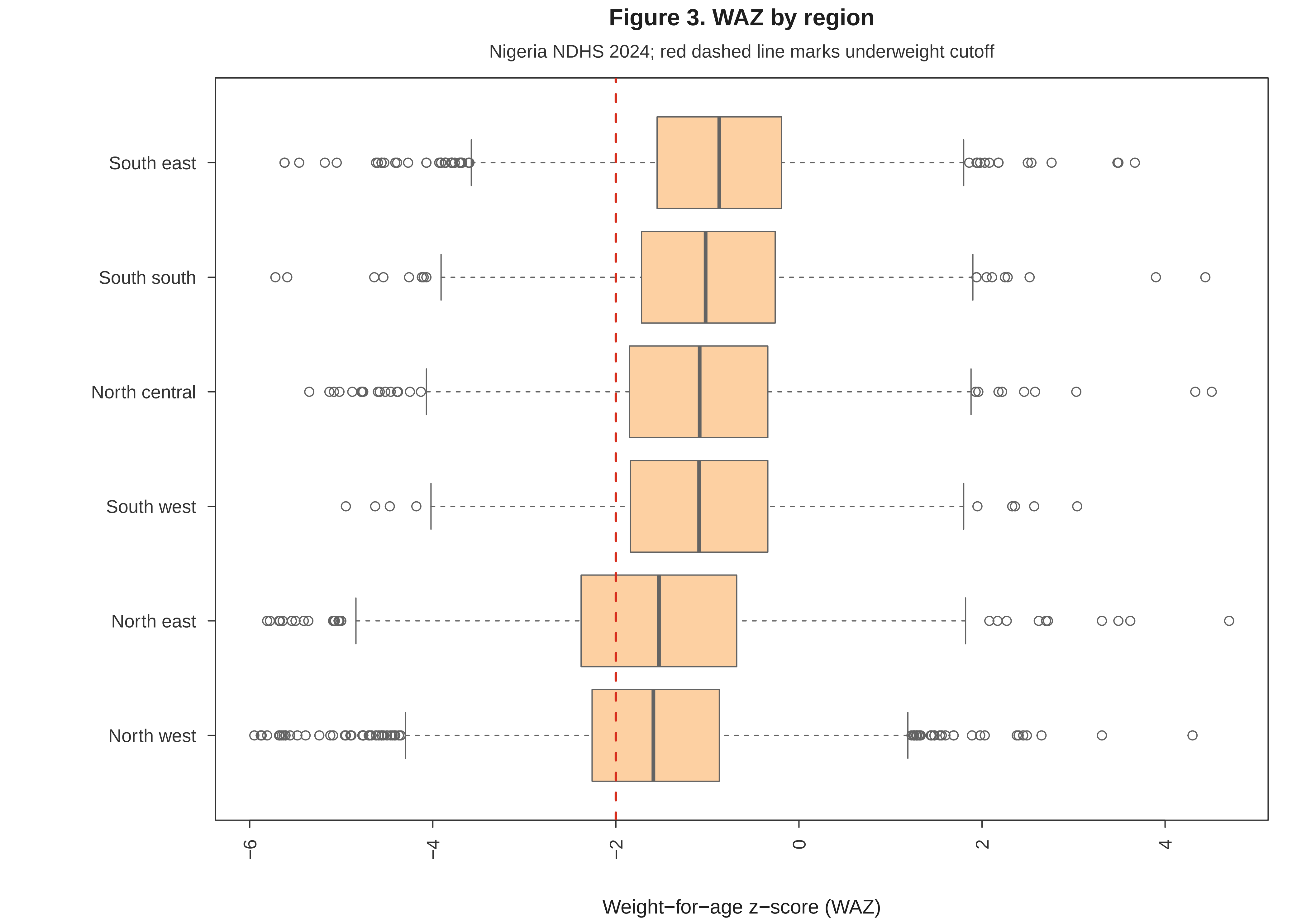
Caption: WAZ distribution with clinical underweight and severe underweight cutoffs.

Figure 2. Distribution of WHZ

Nigeria NDHS 2024; wasting prevalence = 8.4%



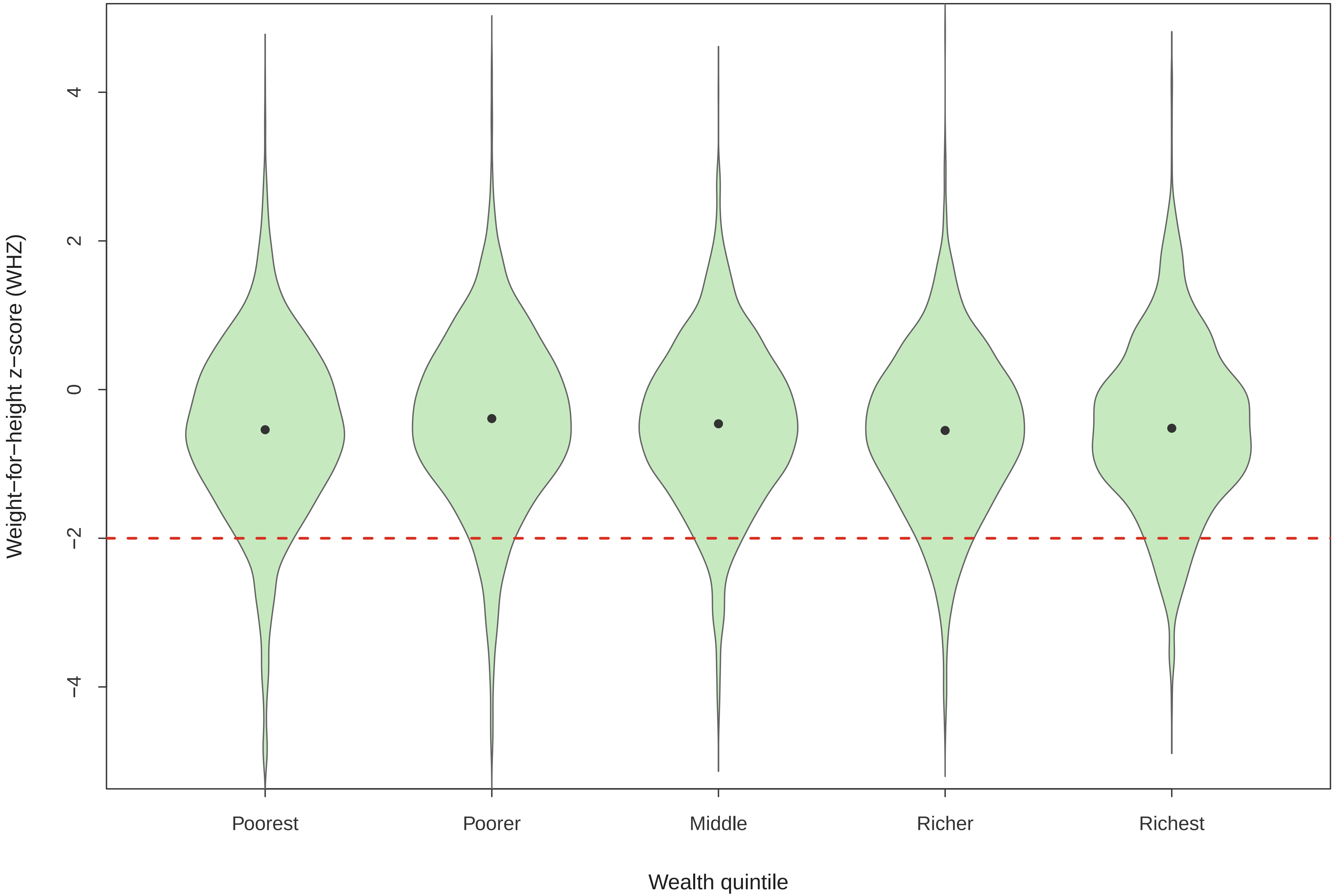
Caption: WHZ distribution with wasting and severe wasting cutoffs.



Caption: Regions are sorted by median WAZ from lowest to highest.

Figure 4. WHZ by wealth quintile

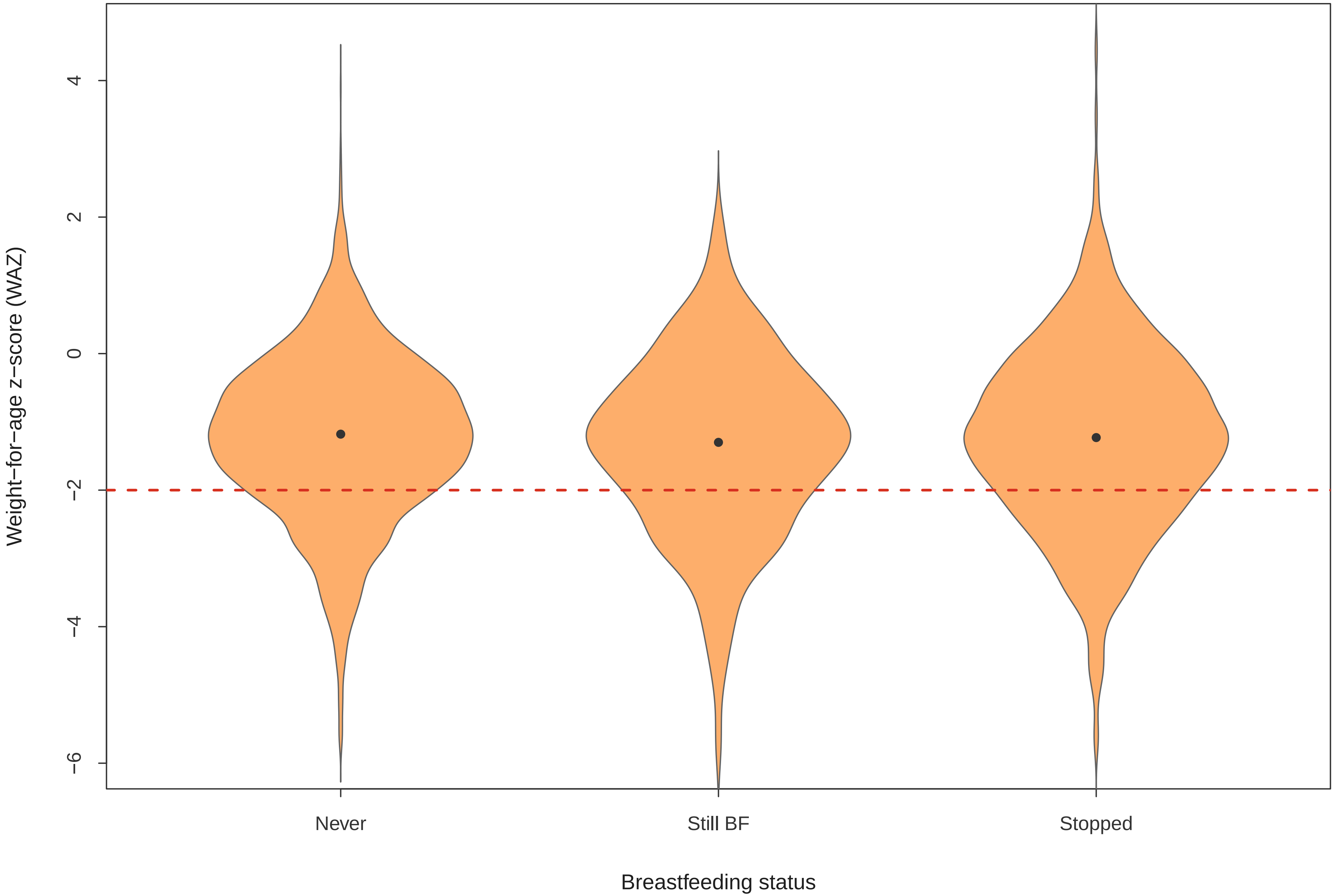
Nigeria NDHS 2024; red dashed line marks wasting cutoff



Caption: WHZ density by household wealth quintile with the wasting cutoff.

Figure 5. WAZ by breastfeeding status

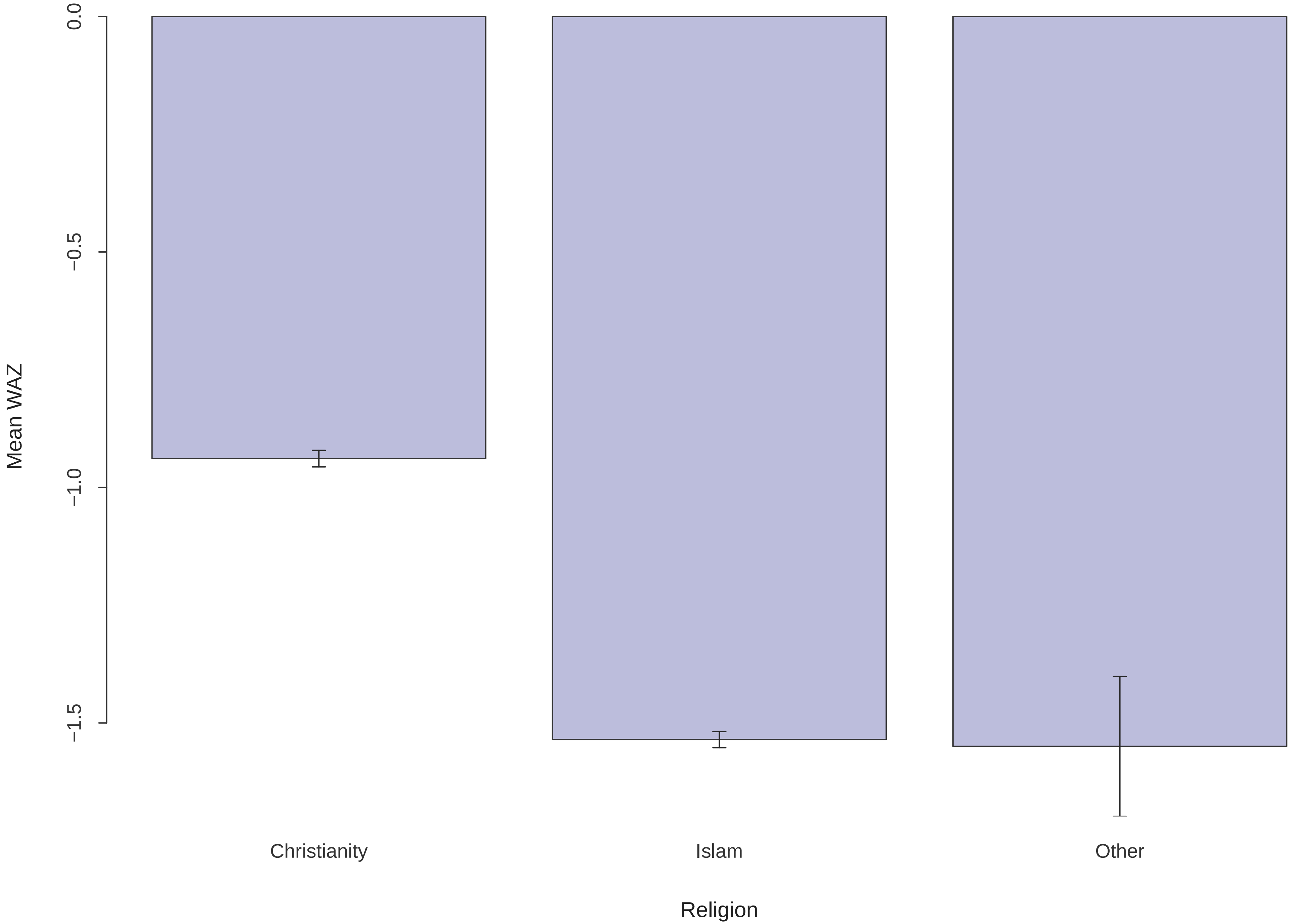
Nigeria NDHS 2024; red dashed line marks underweight cutoff



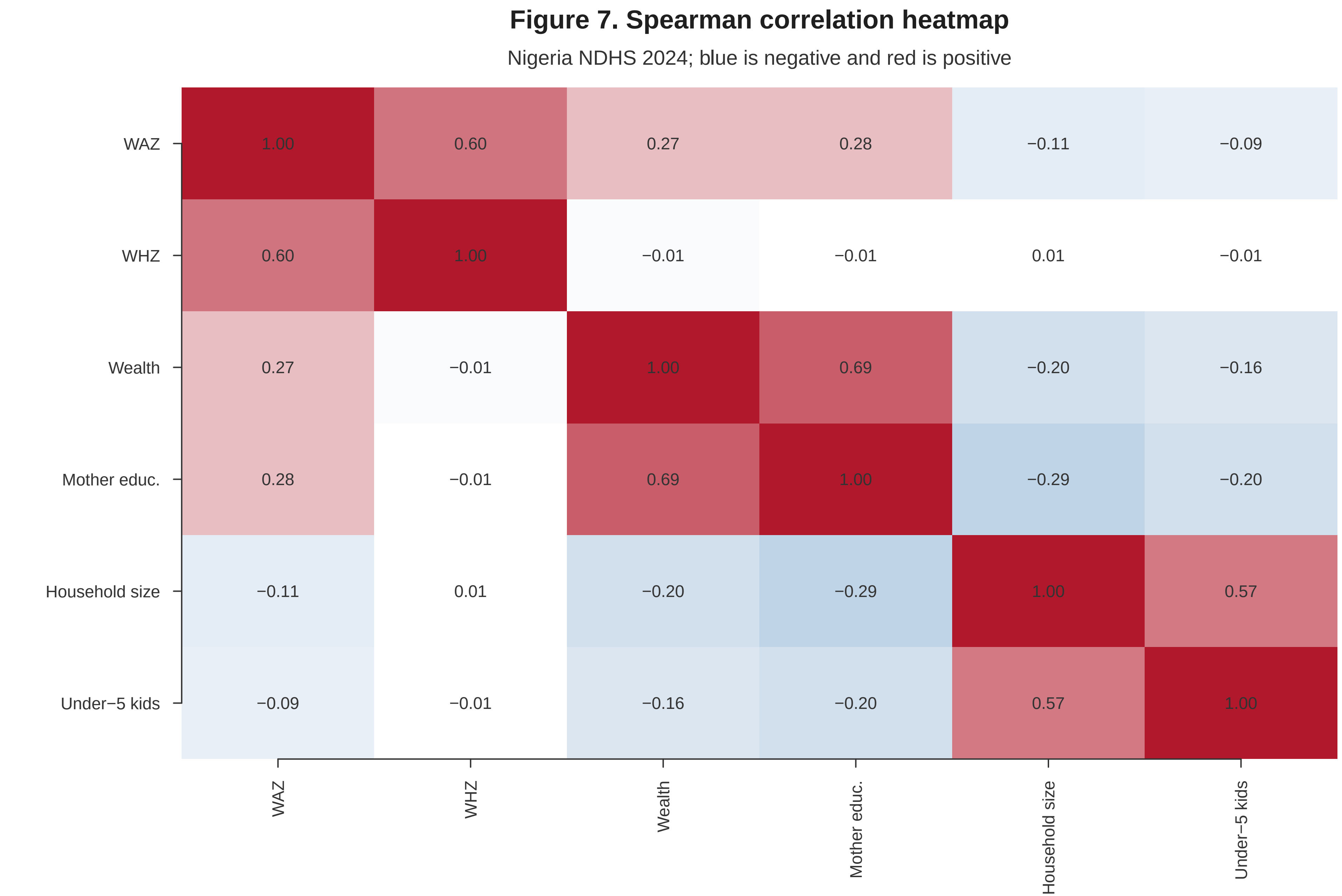
Caption: WAZ density across assignment−defined breastfeeding groups.

Figure 6. Mean WAZ by religion

Nigeria NDHS 2024; error bars show mean +/− standard error



Caption: Mean WAZ by collapsed religion group with standard−error bars.



Caption: Spearman correlation matrix for anthropometry and household predictors.

Task D Correlation and Outliers

Table 6. Ranked Spearman pairs

Variable_1	Variable_2	Spearman_rho	Abs_rho
Wealth	Mothers_Education	0.686	0.686
WAZ	WHZ	0.596	0.596
Household_Size	Under5_Children	0.575	0.575
Mothers_Education	Household_Size	−0.286	0.286
WAZ	Mothers_Education	0.276	0.276
WAZ	Wealth	0.269	0.269
Mothers_Education	Under5_Children	−0.200	0.200
Wealth	Household_Size	−0.195	0.195
Wealth	Under5_Children	−0.157	0.157
WAZ	Household_Size	−0.113	0.113
WAZ	Under5_Children	−0.091	0.091
WHZ	Wealth	−0.012	0.012
WHZ	Household_Size	0.008	0.008
WHZ	Mothers_Education	−0.007	0.007
WHZ	Under5_Children	−0.006	0.006

The strongest non−anthropometric link with WAZ is Mothers_Education; for WHZ it is Wealth. The ranked−pairs table shows that these associations are modest, so correlation should be treated as screening evidence rather than causal evidence.

Table 7. Outlier summary

Variable	Valid_N	IQR_Flagged	IQR_Percent	Z_Flagged	Z_Percent
WAZ	9513	180	1.89	71	0.75
WHZ	9464	184	1.94	79	0.83

Table 8. Outlier decision table

Variable	IQR_Flagged	Percent_Flagged	Main_Regions	Decision
WAZ	180	1.89	North east, North west	Keep
WHZ	184	1.94	North west, North east	Keep

Reason

Anthropometric extremes are plausible child−health findings after DHS invalid codes were removed.
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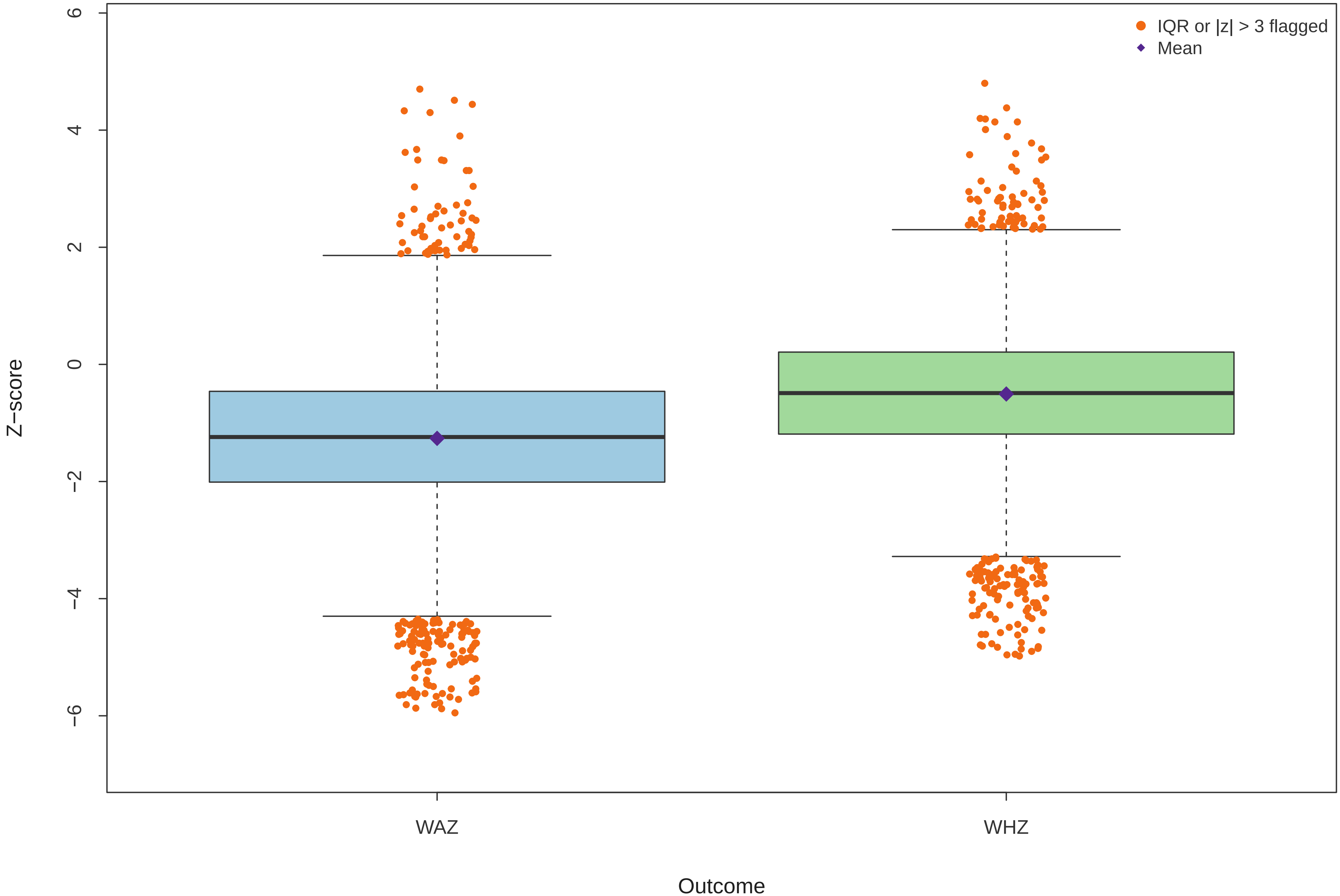
IQR flagged 180 WAZ observations and 184 WHZ observations. These records are kept because the DHS invalid−code rule has already removed non−measurements, and remaining extremes may represent clinically important children.

Figure 8. WAZ and WHZ outlier flags

Nigeria NDHS 2024; orange points are flagged by at least one method



Caption: Side−by−side WAZ and WHZ box plots with orange flagged points and purple means.

Task E Research Questions and Conclusion

Table 9. Future research questions

RQ1

Among children in the North west and North east, is WAZ lower than in southern zones after adjusting for wealth, mother's education, residence, and household size?

Evidence: Table 5 and Figure 3

Suggested method: Weighted linear regression or logistic model for underweight

RQ2

Among children in the poorest wealth quintile, is WHZ lower than in the richer/richest quintiles after controlling for region and mother's education?

Evidence: Figure 4 and Table 5

Suggested method: Multivariable linear regression for WHZ and logistic model for wasting

RQ3

Among Muslim children, is WAZ lower than among Christian children after adjusting for region, wealth, mother's education, and rural residence?

Evidence: Table 5 and Figure 6

Suggested method: Multivariable regression or adjusted logistic model for underweight

Conclusion

The NDHS 2024 child−recode analysis shows that undernutrition in Nigeria is broader than one anthropometric measure. After removing DHS invalid anthropometry codes, 9,513 children had valid WAZ and 9,464 had valid WHZ. The weighted national prevalence of underweight was 26.4%, compared with 8.4% for wasting. Severe underweight affected 8.3%, while severe wasting affected 1.9%. When compared with the HAZ result from class, stunting was 39.1% and remained the largest deficit, consistent with chronic deprivation being more widespread than acute wasting. The most at−risk subgroup in the descriptive tables was Poorest (Wealth quintile), where underweight reached 38.6%. The regional and socioeconomic figures also show that risk is not evenly distributed across Nigerian children, so national averages hide important local vulnerability. Correlations with wealth, education, household size, and under−five crowding were modest, but they identify plausible predictors for the next sprint. The next analytical step should be a weighted multivariable model for underweight and wasting that adjusts for region, wealth, maternal education, household size, and child age.